Model Question Paper: Science and General Studies

Time: 3 Hours Maximum Marks: 28

Instructions: All questions are compulsory. In Section A each question carries 1 mark.

Section A

1. A train 375 m long passes a pole in 15 seconds. What is the speed of the train?

(a) 40 m/s (b) 20 m/s

(c) 30 m/s (d) 25 m/s

2. A car travels 160 km in 4 hours. What is its average speed?

(a) 64 km/h (b) 48 km/h

(c) 40 km/h (d) 24 km/h

3. Which is the largest organ of the human body?

(a) Liver (b) Heart

(c) Skin (d) Lungs

4. A car travels 120 km in 3 hours. What is its average speed?

(a) 16 km/h (b) 48 km/h

(c) 40 km/h (d) 64 km/h

5. An object of mass 1600 kg has a volume of 2 cubic metres. What is its density?

(a) 1120 kg per cubic metre (b) 320 kg per cubic metre

(c) 800 kg per cubic metre (d) 640 kg per cubic metre

6. An object of mass 4500 kg has a volume of 5 cubic metres. What is its density?

(a) 900 kg per cubic metre
(b) 720 kg per cubic metre
(c) 1440 kg per cubic metre
(d) 360 kg per cubic metre

7. A train 180 m long passes a pole in 9 seconds. What is the speed of the train?

(a) 20 m/s
(b) 28 m/s
(c) 16 m/s
(d) 8 m/s

8. A 16 V battery is connected across a resistor of 8 ohm. What current flows through the resistor?

(a) 1 A
(b) 3 A
(c) 4 A
(d) 2 A

9. In which year was the Constitution of India adopted?

(a) 1947
(b) 1949
(c) 1950
(d) 1948

10. How many days are there in a leap year?

(a) 364
(b) 365
(c) 366
(d) 367

11. A 18 V battery is connected across a resistor of 6 ohm. What current flows through the resistor?

(a) 4 A
(b) 3 A
(c) 1 A
(d) 5 A

12. A car travels 150 km in 5 hours. What is its average speed?

(a) 24 km/h
(b) 12 km/h
(c) 18 km/h
(d) 30 km/h

13. Which gas do plants absorb during photosynthesis?

(a) Oxygen (b) Hydrogen

(c) Carbon dioxide (d) Nitrogen

14. The average of 10 numbers is 32. If the number 48 is removed, what is the average of the rest?

(a) 36 (b) 12

(c) 30 (d) 48

15. Who wrote the national anthem of India?

(a) Mahatma Gandhi (b) Sarojini Naidu

(c) Rabindranath Tagore (d) Bankim Chandra Chatterjee

Section B

16. Write three features of the Indian Constitution. [5 marks]

17. Describe two causes of soil erosion. [3 marks]

18. State Ohm's law and write its formula. [5 marks]